Faculty of Computing & Information Technology

Department of Data Science

<u>Natural Language Processing</u>

BS-DS Fall 2023

Instructions

- Submit your work as
  - i) A Jupyter Notebook (.ipynb) with code and outputs
  - ii) File name format: RollNo_Assignment2.ipynb

Topic: Tokenization, Stemming, Lemmatization

Dataset: Amazon Fine Food Reviews (Kaggle)
https://www.kaggle.com/datasets/snap/amazon-fine-food-reviews

Use: `Reviews.csv` (only first **2000 reviews**)

Tools

- `sent_tokenize`
- `word_tokenize`
- `PorterStemmer`
- `WordNetLemmatizer`

Dataset Loading Cell

```
import pandas as pd

df = pd.read_csv("Reviews.csv")

df = df.head(2000)

texts = df["Text"].astype(str)
```

QUESTION 1:

Instructions

1. Perform sentence and word tokenization.
2. Compute:
    - Total tokens (N)
    - Total types (V)
    - Type-Token Ratio (TTR)
3. Print:
    - Top 10 most frequent tokens

Questions (Answer only from output)

1. What is the total number of tokens?
2. What is the vocabulary size?
3. Is TTR closer to 0 or 1?
4. What is the most frequent token?

QUESTION 2:

Instructions

1. Tokenize the corpus using word_tokenize.
2. Remove English stopwords.

    - Do NOT remove the following negation words:
      o no
      o not
      o never
      o n't

3. Modify the stopword list accordingly.
4. Compute:

    o Total tokens after stopword removal
    o Total types after stopword removal
    o Type-Token Ratio (TTR)

5. Print:

    o One example review before stopword removal
    o Same review after stopword removal
    o One example where negation is preserved in the final token list

Questions

6. Does token count decrease after stopword removal? (Yes/No based on output)
7. Does vocabulary size decrease? (Yes/No based on output)
8. Is TTR higher or lower compared to the original corpus?
9. Are negation words still present in the processed tokens? (Yes/No based on output)

QUESTION 3:

Instructions

10. Apply Porter Stemming to all tokens.
11. Compute:
    - Vocabulary size after stemming
    - % reduction in vocabulary
12. Print:
    - 10 examples where multiple words map to same stem

Questions

1. What is the vocabulary size after stemming?
2. What is the percentage reduction?
3. Does stemming reduce vocabulary size significantly? (Yes/No based on numbers)

QUESTION 4:

Instructions

1. Apply WordNetLemmatizer with `pos='v'`
2. Compute:
    - Vocabulary size after lemmatization
3. Print:
    - 10 words where stem ≠ lemma

Questions

1. What is vocabulary size after lemmatization?
2. Is vocabulary after lemmatization larger or smaller than after stemming?
3. How many differences are observed between stem and lemma (from printed examples)?

QUESTION 5:

Instructions

1. Tokenize corpus.
2. Apply stemming.
3. Compute:

Metric	Original	Stemmed
Tokens	?	?
Types	?	?
TTR	?	?

Questions

1. Does token count change after stemming?
2. Does TTR increase or decrease?
3. By how many types did vocabulary decrease?

QUESTION 6:

Instructions

1. Tokenize corpus.
2. Apply lemmatization (`pos='v'`)
3. Compute:

Metric	Original	Lemmatized
Tokens	?	?
Types	?	?
TTR	?	?

Questions

1. Does token count change after lemmatization?
2. Is vocabulary reduction greater in stemming or lemmatization?
3. Which method produces smaller TTR?

QUESTION 7:

Instructions

Select 3 reviews from the first 2000 reviews and show full output at every stage below:

For each selected review, display:

1. Original review
2. After HTML removal
3. After punctuation normalization (!!! → !, ??? → ?, ... → .)
4. After lowercasing
5. After tokenization
6. After stopword removal (negation preserved)
7. After lemmatization (pos='v')

Display full text/tokens at every step. Do not skip any stage.

Questions

1. At which stage does the text length reduce the most?
2. Which step changes the vocabulary structure more significantly — stopword removal or lemmatization?
3. Does the meaning of the review remain understandable after all preprocessing steps? (Yes/No based on observation)